Provincial Agricultural Drought Mitigation Planning: Optimal LDRRMF Allocation for Cavite Province

Jose Aries E. De Los Santos | *jedelossantos1@up.edu.ph*

1 Introduction

Local governments face a recurring challenge during dry seasons: distributing limited disaster funds before drought damages crops. In Cavite, prolonged dry spells driven by the El Niño–Southern Oscillation (ENSO) reduce root-zone soil moisture across agricultural areas, threatening high-value crops such as co□ee and cacao [3, 6]. In practice, municipal disaster resource allocations often rely on uniform across-the-board splits or subjective assessments rather than measured environmental risk, leaving the most vulnerable farming communities underfunded.

To address this gap, this study develops a two-stage decision-support pipeline:

1. A machine learning model (Random Forest) trained on NASA POWER satellite weather data [9] predicts dry-season root-zone soil wetness across all 23 local government units (LGUs).
2. A continuous **Linear Program (LP)** solved via the Simplex method [1, 2, 4] allocates the provincial preparedness fund to maximize risk-weighted drought protection, subject to statutory limits, baseline readiness floors, and regional agricultural priorities mandated by national disaster financing guidelines [5, 8].

This formulation replaces subjective budget decisions with a transparent, data-driven optimization model that balances mathematical e□ciency with public finance law.

2 Statement of the Problem

The Provincial Disaster Risk Reduction and Management Council (PDRRMC) of Cavite is preparing its fiscal allocation plan for the peak dry season. The province comprises 23 local government units, consisting of 16 municipalities and 7 component cities [7].

The research uses a two-stage method:

1. A Random Forest model predicts dry-season root-zone soil wetness ($\widehat{\text{GWETROOT}}_i \in [0,1]$) for each municipality using NASA POWER climate data.

   Because weather data during dry months might be nongaussian and sudden rain showers can make an average misleading, we do not use the simple mean. Instead, we take the 10th percentile (the driest 10% of days) to capture severe drought conditions:

   $$R_i = 1 - Q_{0.10}(\widehat{\text{GWETROOT}}_i) = Q_{0.90}(R_i), \quad \forall i \in \{1, 2, \ldots, 23\}. \tag{1}$$

   Here, the driest 10% of soil moisture maps directly to the top 90% of drought risk (R_i).
2. A linear programming model then distributes the provincial preparedness budget (x_i) to maximize total drought protection, while following government budget limits and minimum funding rules for every town.

The optimization model follows two types of rules: **national legal mandates** (from RA 10121) and **project benchmark parameters** (realistic example values chosen for this case study).

1. **Legal Minimum for Local Calamity Funds (Statutory Law):** Under **Republic Act No. 10121, Section 21, Paragraph 1** c[8], every local government unit (LGU) must set aside at least five percent (5%) of its regular yearly revenue for the Local Disaster Risk Reduction and Management Fund (LDRRMF, denoted as T).

2. **The 30% Quick Response Fund Reserve (Statutory Law):** Under

3 Tasks

As quantitative policy consultants to the Provincial Government of Cavite, you are tasked with completing the following deliverables:

1. **Mathematical Formulation:** Formally define the continuous Linear Programming Problem. Clearly label decision variables, the risk-weighted objective function, and express all policy constraints in standard mathematical inequality notation. Show how the 30% Quick Response Fund (Q) is explicitly partitioned under RA 10121, Section 21.

1. Statutory Fund Partition and Parameter Definitions

In strict adherence to Section 21 of Republic Act No. 10121 and Joint Memorandum Circular No. 2013-1, the benchmark Local Disaster Risk Reduction and Management Fund ($T = \text{PHP } 50{,}000{,}000$) is partitioned into two statutory components prior to optimization:

$$Q = 0.30\,T = \text{PHP } 15{,}000{,}000 \quad \text{(Quick Response Fund, stand-by relief reserve)} \tag{6}$$

$$B = 0.70\,T = \text{PHP } 35{,}000{,}000 \quad \text{(Statutory Pre-Disaster Preparedness Budget)} \tag{7}$$

Because Q is legally encumbered for post-disaster response during a declared State of Calamity, only the preparedness ceiling B is available for proactive allocation.

Let the operational parameters and regional topology of Cavite be defined as follows:

- $\mathcal{M} = \{1, 2, \ldots, 23\}$: The index set denoting all 23 local government units (16 municipalities and 7 component cities) of Cavite ($|\mathcal{M}| = 23$).
- $\mathcal{U} \subset \mathcal{M}$: The subset containing the nine (9) designated upland agricultural LGUs ($|\mathcal{U}| = 9$), leaving 14 lowland LGUs in $\mathcal{M} \setminus \mathcal{U}$.
- $d_i \in \{0, 1\}$: A deterministic parameter encoding regional agricultural zoning:

$$d_i = \begin{cases} 1, & \text{if } i \in \mathcal{U} \\ 0, & \text{if } i \in \mathcal{M} \setminus \mathcal{U} \end{cases}$$

- $R_i \in [0, 1]$: The empirical drought risk score of LGU i, defined by the complement of predicted root-zone soil wetness: $R_i = 1 - \overline{\text{GWETROOT}}_i$.
- $B = \text{PHP } 35{,}000{,}000$: The total provincial statutory preparedness budget ceiling.
- $L_i = \text{PHP } 400{,}000$: The mandatory operational survival floor for each LGU $i \in \mathcal{M}$.
- $U_i = 0.15\,B = \text{PHP } 5{,}250{,}000$: The individual equity absorption ceiling for each LGU $i \in \mathcal{M}$.

2. Primal Problem Formulation

The resource distribution model is formulated as the following continuous Linear Programming Prob-

lem (LPP):

$$
\begin{aligned}
\textbf{maximize} \quad & Z = \sum_{i \in \mathcal{M}} R_i x_i \\
\textbf{subject to:} \quad & \sum_{i \in \mathcal{M}} x_i \leq B && \text{(Statutory Preparedness Budget Ceiling)} \\
& \sum_{i \in \mathcal{M}} d_i x_i \geq 0.60\, B && \text{(Zonal Upland Agricultural Guarantee)} \\
& x_i \geq L_i, \quad \forall i \in \mathcal{M} && \text{(Operational Survival Floor)} \\
& x_i \leq U_i, \quad \forall i \in \mathcal{M} && \text{(Equity Absorption Ceiling)} \\
& x_i \geq 0, \quad \forall i \in \mathcal{M} && \text{(Non-Negativity)}
\end{aligned}
\tag{8}
$$

where each decision variable $x_i \in \mathbb{R}_{\geq 0}$ denotes the continuous fiscal appropriation granted to municipality or city i.

3. Reduction to Canonical Standard Form

To transform the primal formulation into the canonical standard form ($\min \mathbf{c}^T \mathbf{x}$ subject to $\mathbf{A}\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$), we perform the following transformations:

(a) Convert maximization to minimization by negating the objective coe□cients: $\max Z \equiv -\min(-Z)$.

(b) Introduce slack variable $s_1 \geq 0$ for the statutory budget ceiling constraint.

(c) Introduce surplus variable $e_1 \geq 0$ for the zonal upland guarantee constraint.

(d) Introduce surplus variables $v_i \geq 0$ for each $i \in \mathcal{M}$ for the operational survival floor constraints.

(e) Introduce slack variables $u_i \geq 0$ for each $i \in \mathcal{M}$ for the equity absorption ceiling constraints.

The scalar equality system is expressed as:

$$
\begin{aligned}
\textbf{minimize} \quad & \sum_{i \in \mathcal{M}} -R_i x_i \\
\textbf{subject to:} \quad & \sum_{i \in \mathcal{M}} x_i + s_1 = 35{,}000{,}000 \\
& \sum_{i \in \mathcal{M}} d_i x_i - e_1 = 21{,}000{,}000 \\
& x_i - v_i = 400{,}000, \quad \forall i \in \mathcal{M} \\
& x_i + u_i = 5{,}250{,}000, \quad \forall i \in \mathcal{M} \\
& x_i, v_i, u_i \geq 0 \quad (\forall i \in \mathcal{M}), \quad s_1, e_1 \geq 0
\end{aligned}
\tag{9}
$$

4. Matrix Representation

The system comprises $m = 48$ linear equality constraints and $n = 71$ continuous non-negative variables. The augmented solution vector $\mathbf{x} \in \mathbb{R}^{71}$, cost vector $\mathbf{c} \in \mathbb{R}^{71}$, and requirement vector $\mathbf{b} \in \mathbb{R}^{48}$

are partitioned in block form as:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_{23\times 1} \\ s_1 \\ e_1 \\ \mathbf{v}_{23\times 1} \\ \mathbf{u}_{23\times 1} \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -R_1 \\ \vdots \\ -R_{23} \\ 0 \\ 0 \\ \mathbf{0}_{23\times 1} \\ \mathbf{0}_{23\times 1} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 35{,}000{,}000 \\ 21{,}000{,}000 \\ 400{,}000 \cdot \mathbf{1}_{23\times 1} \\ 5{,}250{,}000 \cdot \mathbf{1}_{23\times 1} \end{bmatrix}$$

The structural constraint matrix $\mathbf{A} \in \mathbb{R}^{48\times 71}$ is assembled as:

$$\mathbf{A} = \begin{bmatrix} \mathbf{1}_{23\times 1}^T & 1 & 0 & \mathbf{0}_{23\times 1}^T & \mathbf{0}_{23\times 1}^T \\ \mathbf{d}_{23\times 1}^T & 0 & -1 & \mathbf{0}_{23\times 1}^T & \mathbf{0}_{23\times 1}^T \\ \mathbf{I}_{23} & \mathbf{0}_{23\times 1} & \mathbf{0}_{23\times 1} & -\mathbf{I}_{23} & \mathbf{0}_{23\times 23} \\ \mathbf{I}_{23} & \mathbf{0}_{23\times 1} & \mathbf{0}_{23\times 1} & \mathbf{0}_{23\times 23} & \mathbf{I}_{23} \end{bmatrix} \tag{10}$$

where $\mathbf{I}_{23}$ is the 23×23 identity matrix, $\mathbf{1}_{23\times 1}$ is a column vector of ones, and $\mathbf{d} = [d_1, \ldots, d_{23}]^T$ is the static upland indicator vector. Because every decision, slack, and surplus variable belongs to $\mathbb{R}_{\geq 0}$, the problem defines a bounded convex polyhedron in $\mathbb{R}^{71}$ solvable via standard continuous Primal Simplex algorithms.

2. **Algorithmic Implementation Formulation:** To execute the optimization numerically via modern solvers such as `scipy.optimize.linprog` (using `method='highs'`), the continuous LPP is cast into the canonical bounded upper-inequality form:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & \mathbf{A}_{\text{ub}}\mathbf{x} \leq \mathbf{b}_{\text{ub}} \\ & \boldsymbol{\ell} \leq \mathbf{x} \leq \mathbf{u} \end{aligned} \tag{11}$$

Rather than augmenting the system with 46 auxiliary slack and surplus variables for individual municipal floors and ceilings, modern implementations handle box constraints directly via variable bounds. This reduces the structural inequality matrix to only $m = 2$ global coupling constraints over $n = 23$ decision variables:

- **Objective Vector ($\mathbf{c} \in \mathbb{R}^{23}$):** Since `linprog` minimizes by default, the risk maximization objective is converted by setting:

$$\mathbf{c} = -\mathbf{R} = -[R_1, R_2, \ldots, R_{23}]^T \tag{12}$$

- **Decoupled Box Bounds ($\boldsymbol{\ell}, \mathbf{u} \in \mathbb{R}^{23}$):** The survival floors and absorption ceilings are isolated from the row constraints and supplied via the `bounds` parameter:

$$\ell_i = L_i = \text{PHP } 400{,}000, \quad u_i = U_i = \text{PHP } 5{,}250{,}000, \quad \forall i \in \mathcal{M} \tag{13}$$

- **Coupling Inequality System ($\mathbf{A_{ub}} \in \mathbb{R}^{2\times 23}$, $\mathbf{b_{ub}} \in \mathbb{R}^2$):** The remaining macro-policy constraints consist of the global statutory budget ceiling and the zonal upland guarantee. Inverting the greater-than inequality of the upland guarantee to conform to the standard $\leq$ format yields:

$$\mathbf{A}_{\text{ub}} = \begin{bmatrix} \mathbf{1}_{23}^T \\ -\mathbf{d}^T \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ -d_1 & -d_2 & \cdots & -d_{23} \end{bmatrix} \in \mathbb{R}^{2\times 23} \tag{14}$$

and the corresponding right-hand side capacity vector:

$$\mathbf{b}_{\text{ub}} = \begin{bmatrix} B \\ -0.60\,B \end{bmatrix} = \begin{bmatrix} 35{,}000{,}000 \\ -21{,}000{,}000 \end{bmatrix} \in \mathbb{R}^{2\times 1} \tag{15}$$

This compact formulation matches the mathematical inputs passed directly to the solver routine while eliminating redundant basis inversions.

3. **Budget Allocation Analysis**

Analysis based on Table 1

(a) Identify which municipalities reached the maximum absorption ceiling (U_i = PHP $5{,}250{,}000$) and which remained at the operational baseline floor (L_i = PHP $400{,}000$).

- **The Maximum Limit (PHP 5,250,000):** Five upland towns (Alfonso, Amadeo, Magallanes, Mendez, and Indang) received the maximum allowed budget. Why? Because our machine learning model identified them as having the highest drought risk (scores around 0.54 to 0.55). We put a "ceiling" (15% maximum) on these towns so that the most at-risk areas get the most help, but they do not drain the entire provincial fund.
- **The Remaining Balance (PHP 1,950,000):** General Emilio Aguinaldo also had a very high risk score (0.528) and was next in line. However, after funding the top five towns and setting aside basic emergency funds for everyone else, the province only had PHP 1,950,000 left. The formula simply gives the exact remaining "change" to this town to completely use up the budget.
- **The Baseline Minimum (PHP 400,000):** The other 17 towns had lower drought risk scores (0.41 and below). Because their agricultural risk is lower this season, they stay at the "floor" limit. This guarantees that every single town gets a basic PHP 400,000 to keep their local disaster o□ces running, ensuring no town is left at zero.

(b) Explain the optimization model's decision-making logic: how did the algorithm balance providing emergency readiness funds to every town, prioritizing the most vulnerable upland farming areas, and preventing any single municipality from receiving an excessive share of public funds?

This is calculated using an algorithm based on George Dantzig's *Simplex Method* [1, 2, 4]. It is highly reliable because it acts like a strict, logical accountant following a clear set of rules:

(a) **Pay the Minimums First:** The math instantly sets aside PHP 400,000 for all 23 towns. This locks up PHP 9.2 million of the 35 million budget right away, ensuring basic survival funds for everyone.

(b) **Rank by Risk:** It takes the remaining PHP 25.8 million and sorts the towns strictly from highest drought risk to lowest.

(c) **Apply the Cap (Ceiling):** It starts pouring money into the highest-risk towns. Once a town hits the 15% legal limit (PHP 5.25 million), the math cuts them o□ and moves to the next town on the list.

(d) **Stop at Zero:** When it reaches the sixth town (General Emilio Aguinaldo), the money runs out. The math stops exactly at the PHP 35 million total.

By using this mathematical method, we can prove to the provincial government that every single peso was assigned based purely on climate data and budget laws, completely removing human bias and politics from the decision.

4. **Scenario Analysis: Climate Stress-Testing under an Intensifying El Niño Advisory**

   Grounded in the official DOST-PAGASA Climate Advisory (Press Release No. 225) [3] warning of an intensifying "Very Strong" El Niño event, conduct a counterfactual sensitivity experiment to evaluate the resilience of the combined machine learning and linear programming allocation pipeline. Rather than manually altering budgetary quotas, apply an empirical climatological perturbation directly to the dry-season evaluation features:

$$\text{PRECTOTCORR}' = 0.70 \times \text{PRECTOTCORR}, \quad \text{T2M}' = \text{T2M} + 1.5^{\circ}\text{C} \tag{16}$$

   representing a 30% reduction in precipitation (consistent with PAGASA's official dry spell / below-normal criteria) coupled with a $+1.5^{\circ}$C surface air temperature anomaly.

Table 1: Optimal LDRRMF Preparedness Allocation via Simplex Method

Municipality	**Zone**	**Risk Score (R_i)**	**Allocated (PHP)**	**% of Budget**
Ceiling Allocations ($U_i =$ PHP 5,250,000)				
Alfonso	Upland	0.549991	5,250,000	15.00
Amadeo	Upland	0.549991	5,250,000	15.00
Magallanes	Upland	0.549560	5,250,000	15.00
Mendez	Upland	0.549311	5,250,000	15.00
Indang	Upland	0.539777	5,250,000	15.00
Marginal Budget-Clearing Allocation				
Gen. Emilio Aguinaldo	Upland	0.528011	1,950,000	5.57
Baseline Floor Allocations ($L_i =$ PHP 400,000)				
Tagaytay City	Upland	0.407163	400,000	1.14
Silang	Upland	0.388931	400,000	1.14
Maragondon	Upland	0.410000	400,000	1.14
Ternate	Lowland	0.410000	400,000	1.14
Tanza	Lowland	0.410000	400,000	1.14
Rosario	Lowland	0.410000	400,000	1.14
Noveleta	Lowland	0.410000	400,000	1.14
Naic	Lowland	0.410000	400,000	1.14
Kawit	Lowland	0.407000	400,000	1.14
Imus City	Lowland	0.409000	400,000	1.14
General Trias	Lowland	0.410000	400,000	1.14
Gen. Mariano Alvarez	Lowland	0.354754	400,000	1.14
Dasmarinas City	Lowland	0.410000	400,000	1.14
Cavite City	Lowland	0.410000	400,000	1.14
Carmona	Lowland	0.380971	400,000	1.14
Bacoor City	Lowland	0.400923	400,000	1.14
Trece Martires City	Lowland	0.410000	400,000	1.14
Total Budget Distributed			**35,000,000**	**100.00**

References

[1] D. Bertsimas and J. N. Tsitsiklis. *Introduction to Linear Programming*. Athena Scientific, Belmont, MA, 1997.

[2] G. B. Dantzig. *Linear Programming and Extensions*. Princeton Landmarks in Mathematics and Physics. Princeton University Press, Princeton, NJ, 1998. ISBN 978-0691059136.

[3] DOST-PAGASA. Climate Advisory: DOST-PAGASA warns of intensifying 'very strong' El ñiño persisting into first half of 2027. O□cial Press Release, Sept. 2026. URL https://pagasa.dost.gov.ph/press-release/225. Published on September 23, 2026.

[4] F. S. Hillier and G. J. Lieberman. *Introduction to Operations Research*. McGraw-Hill Education, Dubuque, IA, 11th edition, 2021. ISBN 9781259872990.

[5] National Disaster Risk Reduction and Management Council (NDRRMC) and Department of Budget and Management (DBM) and Department of the Interior and Local Government (DILG). Joint Memorandum Circular No. 2013-1: Allocation and Utilization of the Local Disaster Risk Reduction and Management Fund (LDRRMF), Mar. 2013.

[6] Provincial Government of Cavite. Cavite Provincial Development and Physical Framework Plan (PDPFP) 2021–2030. Technical report, Provincial Planning and Development O□ce, Trece Martires City, Cavite, Philippines, 2021.

[7] Provincial Government of Cavite. Cavite Provincial Development and Physical Framework Plan (PDPFP) 2021–2030. Provincial framework plan, O□ce of the Provincial Planning and Development Coordinator, Trece Martires City, Cavite, Philippines, 2021. URL https://cavite.gov.ph/wp-content/uploads/2026/05/CAVITE_PDPFP.pdf. Accessed via the o□cial Provincial Government of Cavite portal.

[8] Republic of the Philippines. Republic Act No. 10121: Philippine Disaster Risk Reduction and Management Act of 2010, Section 21, Paragraph 1. O□cial Gazette, May 2010. URL https://lawphil.net. Approved on May 27, 2010.

[9] A. H. Sparks. nasapower: A NASA POWER Global Meteorology, Surface Solar Energy and Climatology Data Client. *The Journal of Open Source Software*, 3(30):1035, 2018. doi: 10.21105/joss.01035.